# Monthly Huabei Loan Sales Monitoring Dashboard

## 1. Project Overview

This project focused on building an automated business intelligence dashboard for monitoring monthly Huabei loan installment sales performance across regions and product terms. The dashboard was developed using Tableau and Excel-based data pipelines to support regional sales management and operational decision-making.

The solution consolidated transaction-level sales data into an interactive analytical platform that enabled stakeholders to monitor KPIs, compare regional performance, analyze customer trends, and evaluate installment product effectiveness in real time.

The dashboard supported approximately $97M in monthly loan transactions and was designed for regional managers, operations teams, and executive stakeholders.

# 2. Business Background

Ant Group's installment loan business generates large volumes of transaction data across multiple provinces and installment products. Prior reporting processes relied heavily on manually maintained Excel reports, creating several operational challenges:

- Time-consuming monthly reporting workflows
- Inconsistent KPI calculations across teams
- Limited visibility into regional sales performance
- Difficulty identifying product-level growth trends
- Delayed business decision-making

The primary objective of this project was to create a centralized and automated reporting solution capable of:

- Monitoring sales and customer performance across regions
- Tracking month-over-month (MoM) growth
- Comparing installment product performance
- Supporting strategic sales planning
- Reducing manual reporting effort

# 3. Data Sources & Preparation

## Data Sources

The dashboard integrated multiple Excel-based datasets containing:

| Dataset | Description |
|---|---|
| Transaction Data | Monthly loan transaction records |
| Regional Mapping | Province-to-region mapping |
| Product Information | Installment term categories |
| KPI Metrics | Sales and customer indicators |

## Data Preparation Process

Data cleaning and transformation were performed using Tableau Prep and Excel Power Query.

### Key preprocessing steps included:

- Standardizing province and region naming conventions
- Cleaning missing or inconsistent transaction records
- Formatting and validating date fields
- Aggregating metrics by region and installment term
- Removing duplicate records
- Creating calculated KPI fields

The workflow enabled automated refreshes for recurring monthly reports and improved reporting consistency across business teams.

# 4. KPI Design

The dashboard focused on four primary business KPIs:

| KPI | Business Purpose |
| --- | --- |
| Sales Amount | Measures total transaction volume |
| Customer Count | Tracks customer participation |
| Average Order Value (AOV) | Evaluates transaction quality |
| MoM Growth | Measures monthly performance changes |

### AOV Formula

$$AOV = \frac{Sales\ Amount}{Customer\ Count}$$

### MoM Growth Formula

$$MoM\ Growth = \frac{Current\ Month - Previous\ Month}{Previous\ Month} \times 100\%$$

Additional calculated metrics were created for:

- Regional contribution percentage
- Product term sales proportion
- Customer growth rate
- Product-level AOV comparison

# 5. Dashboard Design & Visualization Logic

The dashboard was designed with a layered analytical structure to support both executive-level monitoring and detailed operational analysis.

## Top-Level KPI Summary

The top section displayed high-level KPIs including:

- Total Sales Amount
- Customer Count
- Average Order Value
- Overall MoM performance

This enabled stakeholders to quickly assess overall business health.

## Regional & Provincial Analysis

Interactive bar charts and comparison tables were used to analyze:

- Regional sales distribution
- Province-level performance
- Customer growth by region
- AOV differences across provinces

This section helped identify high-performing and underperforming markets.

## Trend Analysis

Line charts were implemented to monitor:

- Daily regional sales trends
- Product term transaction changes
- Short-term performance fluctuations

Trend analysis supported operational forecasting and anomaly detection.

## Product Term Analysis

The dashboard compared installment products across:

- Sales contribution
- Customer volume
- Growth performance
- Average transaction value

This helped evaluate which installment terms generated the strongest business performance.

# 6. Technical Implementation

## Tools & Technologies

| Tool | Purpose |
| --- | --- |
| Tableau Desktop | Dashboard development |
| Tableau Prep | Data cleaning and transformation |

| Tool | Purpose |
|---|---|
| Excel | Source data management |
| Power Query | Automated preprocessing |

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## Key Tableau Features Used

- Calculated Fields
- Interactive Filters
- Dashboard Actions
- Parameters
- Custom KPI Cards
- Trend Visualizations
- Dynamic Region Filtering

## Automation Improvements

The reporting pipeline automated repetitive reporting tasks through Tableau extract refreshes and Power Query preprocessing workflows.

### Results

- Reduced manual reporting effort by approximately 30%
- Improved reporting consistency
- Accelerated monthly reporting delivery
- Enhanced stakeholder accessibility to KPIs

# 7. Key Business Insights

Several important insights were identified through the dashboard analysis:

## 1. Eastern region generated the highest transaction volume

The East region consistently contributed the largest share of total sales and customer activity.

## 2. Long-term installment products drove higher sales amounts

12-term and 18-term installment products generated significantly larger transaction volumes compared to shorter-term products.

## 3. Customer growth and AOV did not always move together

Certain provinces showed strong customer growth while experiencing declining AOV, suggesting expansion into lower-value customer segments.

## 4. Regional performance volatility varied significantly

Northwest regions demonstrated greater month-over-month fluctuations compared to more stable eastern markets.

# 8. Challenges

Several challenges were encountered during development:

- Maintaining consistency across multiple Excel source files
- Designing compact dashboards with large amounts of information
- Balancing dashboard interactivity with performance optimization
- Standardizing KPI definitions across regions

These challenges were addressed through preprocessing automation and standardized metric calculations.

# 9. Future Improvements

Potential future enhancements include:

- SQL-based data warehouse integration
- Real-time data refresh pipelines
- Predictive sales forecasting models
- Customer segmentation analysis
- Mobile dashboard optimization
- Risk and delinquency analytics integration

# 10. Conclusion

This project demonstrated the development of a scalable and automated Tableau reporting solution for financial sales analytics. By integrating data cleaning workflows, KPI standardization, and interactive visualizations, the dashboard improved reporting efficiency and enabled more data-driven operational decision-making.

The project strengthened practical experience in:

- Business intelligence reporting
- Tableau dashboard development
- KPI design
- Data preprocessing automation
- Cross-functional business analytics